

CLUSTER PROFILING & BUSINESS SEGMENTATION.

CUSTOMER SEGMENTATION — FROM TWO CLUSTERS TO BUSINESS PERSONAS

Demographic and spending analysis, shopping-channel behavior, marketing response, named business segments, customer personas, and final visualizations — profiling a 2-cluster K-Means solution for business use.



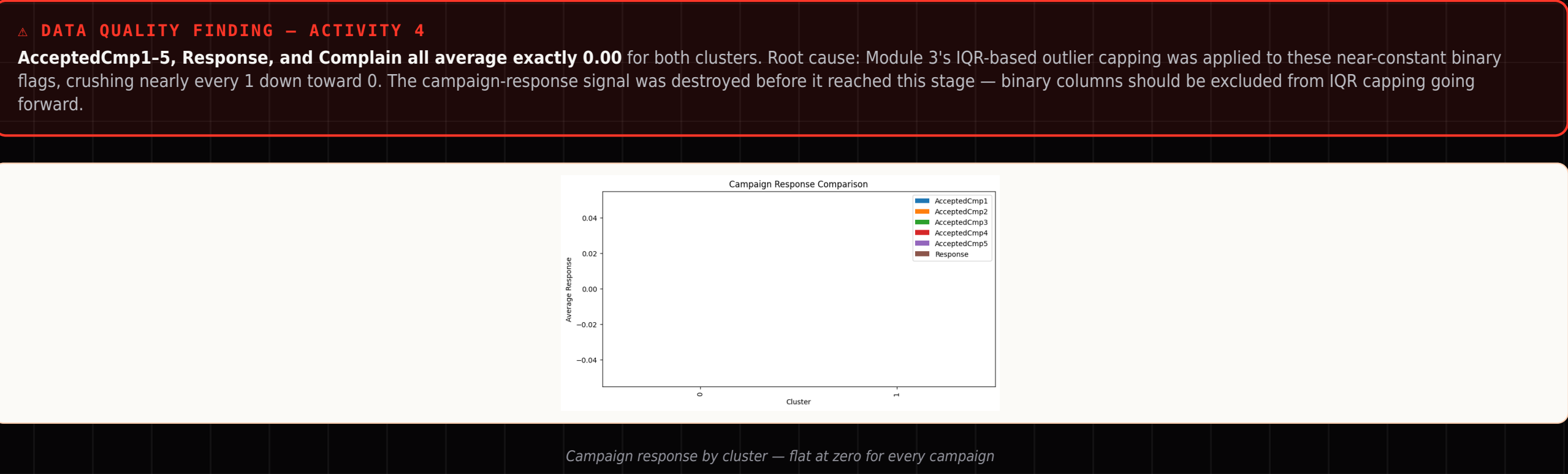
DATASET	2,038 rows × 42 cols (scaled)	CLUSTERS FOUND	2	CLUSTER 0	Premium Loyal — 1,111 (54.5%)	CLUSTER 1	Digital-First* — 927 (45.5%)
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01 — CLUSTER COMPARISON

z-scores

Metric	Cluster 0	Cluster 1
Income	+0.67	−0.80
Total Spending	+0.81	−0.97
Customer Age	+0.18	−0.20
Family Size	−0.32	+0.38
Web Purchases	+0.66	−0.78
Store Purchases	+0.71	−0.86
Catalog Purchases	+0.73	−0.88
Website Visits	−0.40	+0.47
Deal Dependency	−0.60	+0.72

Values are standardized (z-score) units, not real dollars/counts — directional comparison only.



TOP SEGMENT

PREMIUM LOYAL CUSTOMERS

Cluster 0 · higher income, higher spending on every product, more purchases on every channel. Persona: Sarah, 45-60, high income, small family. VIP rewards & loyalty programs.

1,111

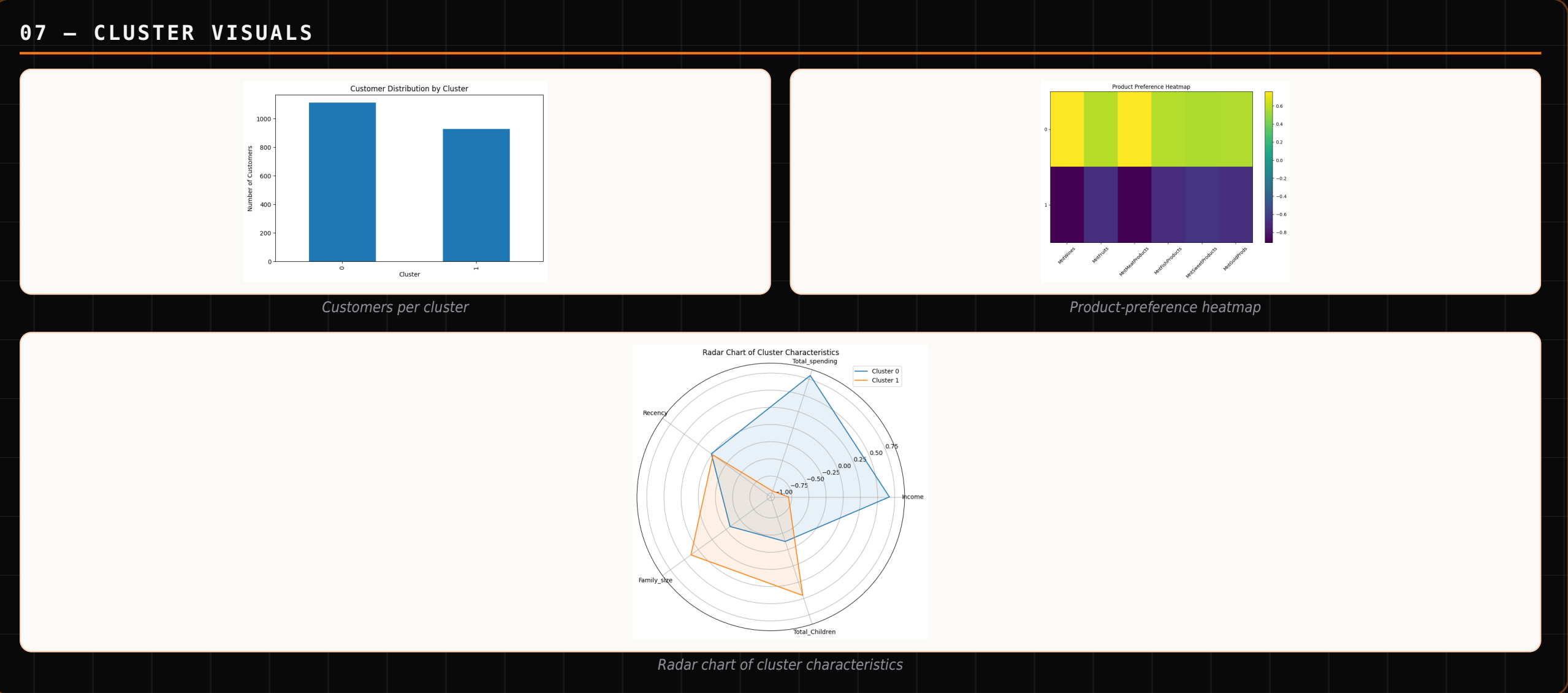
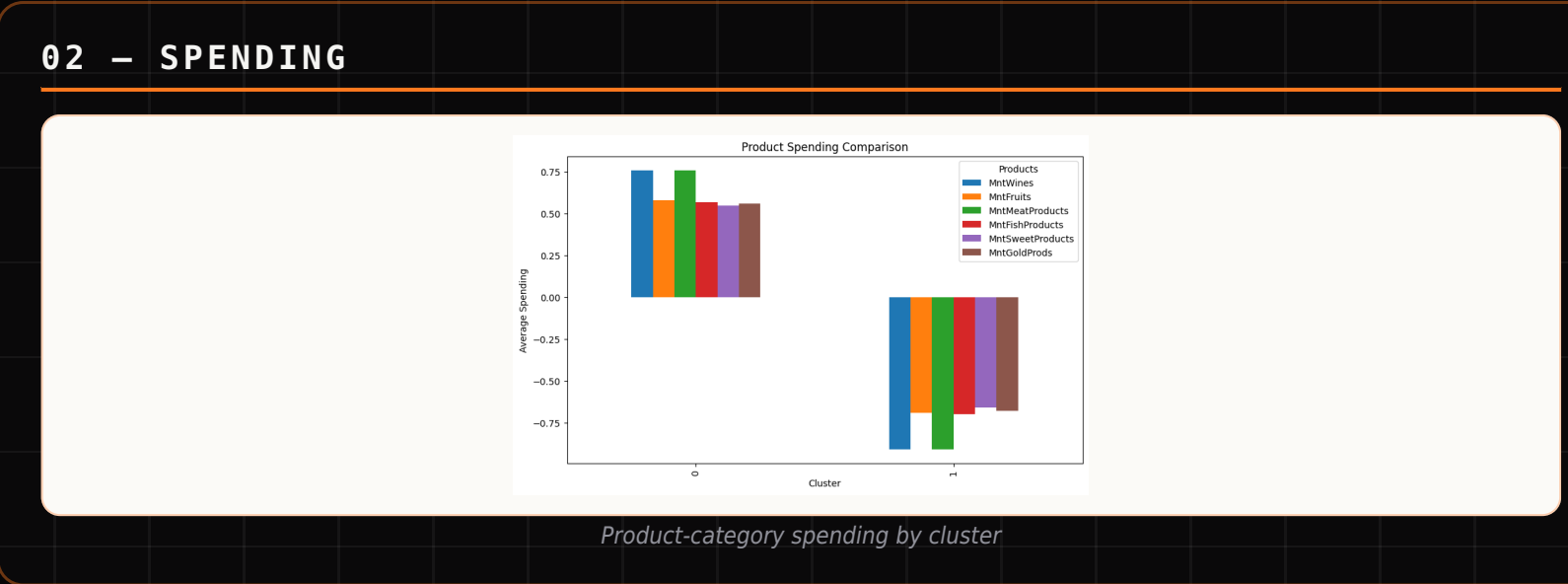
CUSTOMERS

+0.81

SPENDING

Very High

EST. CLV



05 — BUSINESS SEGMENTS

Segment	Persona	Data-Backed?
Premium Loyal Customers	Sarah	Yes
Digital-First Buyers*	Ali	Yes*
Discount-Driven Families	Ahmed	No
At-Risk Customers	John	No

*Named "Digital-First" but shows the **lowest** purchases on every channel — only web *visits* are higher. Only 2 of 4 segments exist in the current model.

- 06 — KEY LIMITATIONS
- **Z-scores, not real units:** Income/spending figures need inverse-transforming before sharing externally.
 - **Campaign signal destroyed:** re-run outlier capping excluding binary flag columns.
 - **Segment count mismatch:** 4 personas defined, only 2 clusters exist — re-cluster at k=4 or drop 2 segments.
 - **Cluster 1 naming:** "Digital-First" doesn't match its lowest-purchase-on-every-channel profile.